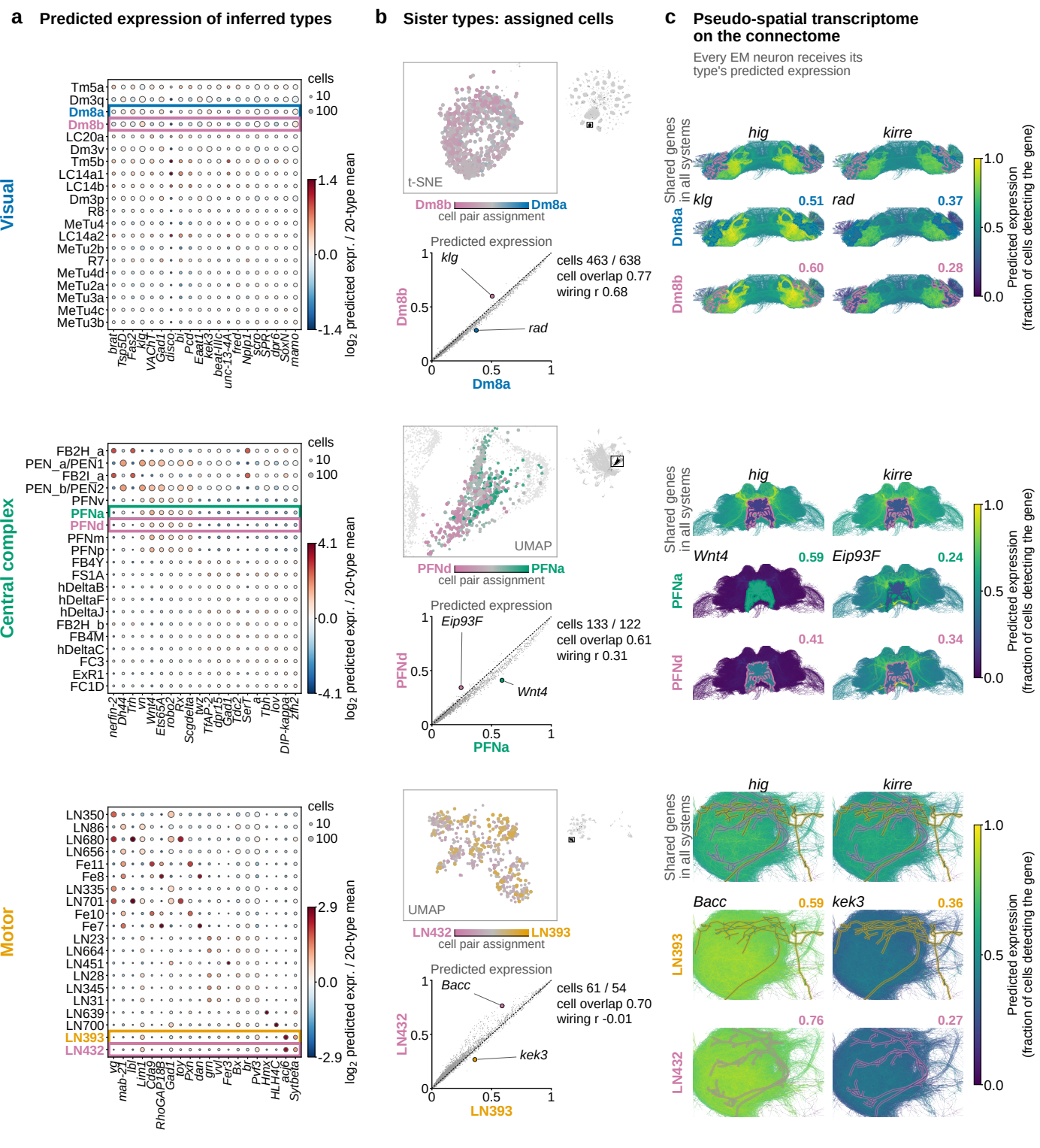

a
Predicted expression of inferred types
b
Sister types: assigned cells
c
Pseudo-spatial transcriptome on the connectome
Every EM neuron receives its type's predicted expression
Visual
Central complex
Motor
cells
10
100
log2 predicted expr. / 20-type mean
Tm5a
Dm3q
Dm8a
Dm8b
LC20a
Dm3v
Tm5b
LC14a1
LC14b
Dm3p
R8
MeTu4
LC14a2
MeTu2b
R7
MeTu4d
MeTu2a
MeTu3a
MeTu4c
MeTu3b
1.4
0.0
-1.4
brat
Tsp5D
Fas2
klg
VAChT
Gad1
disco
bi
Pcd1
Eaat3
kek3
beat-IIIc
unc-13-4A
fred
Nplp1
scrg
SPR
dpr6
SoxN
mamo
t-SHE
Dm8b
Dm8a
cell pair assignment
Predicted expression
klg
rad
cells 463 / 638
cell overlap 0.77
wiring r 0.68
Dm8b
Dm8a
Shared genes in all systems
hig
kirre
klg
0.51
rad
0.37
0.60
0.28
Predicted expression (fraction of cells detecting the gene)
1.0
0.5
0.0
FB2H_a
PEN_a/PEN1
FB2I_a
PEN_b/PEN2
PFNv
PFNa
PFNd
PFNm
PFNp
FB4Y
FS1A
hDeltaB
hDeltaF
hDeltaJ
FB2H_b
FB4M
hDeltaC
FC3
ExR1
FC1D
4.1
0.0
-4.1
nerfin-2
Dh44
Trh
vn
Wnt4
Ets65A
robo2
Rx
Scgdelta
eIF4E1
TfAP-2
dpr15
Gad1
Tdc2
SerT
a
Tbh
lov
DIP-kappa
Zfh2
UMAP
PFNd
PFNa
cell pair assignment
Predicted expression
Eip93F
Wnt4
cells 133 / 122
cell overlap 0.61
wiring r 0.31
PFNd
PFNa
Shared genes in all systems
hig
kirre
Wnt4
0.59
Eip93F
0.24
0.41
0.34
Predicted expression (fraction of cells detecting the gene)
1.0
0.5
0.0
LN350
LN86
LN680
LN656
Fe11
Fe8
LN335
LN701
Fe10
Fe7
LN23
LN664
LN451
LN28
LN345
LN31
LN639
LN700
LN393
LN432
2.9
0.0
-2.9
vg
mab-21
lbl
Lim1
Cdag
RhoGAP18B
Gad1
lov
Pax
dan
dan
vvl
Fer3
Bx
bi
Pvr3
Hmx
HLH4C
ato
Syt6eta
UMAP
LN432
LN393
cell pair assignment
Predicted expression
Bacc
kek3
cells 61 / 54
cell overlap 0.70
wiring r -0.01
LN432
LN393
Shared genes in all systems
hig
kirre
Bacc
0.59
kek3
0.36
0.76
0.27
Predicted expression (fraction of cells detecting the gene)
1.0
0.5
0.0